

A Fast Algorithm for Generating All Triangulations of Biconnected Plane Graphs

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Overview

- 1 Introduction
- 2 Algorithm for Cycle Graphs
- 3 Graph Examples

The Problem:

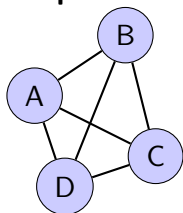
- Generate **all possible triangulations** of biconnected plane graphs
- Previous methods fail with complex structures due to multi-edge creation
- Important applications in graphics, engineering, and chip design

Our Contribution:

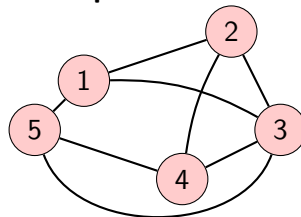
- Novel algorithm with **constant time per triangulation**
- Solves the multi-edge problem using **safe root** concept
- Practical for real-world applications

Example Graphs

Simple Biconnected Graph



Complex Graph with Curved

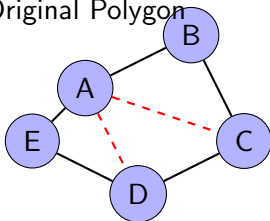


Edges

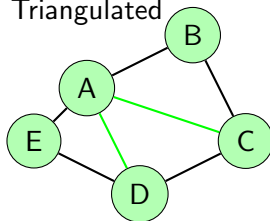
Triangulation Process

Step-by-step triangulation example:

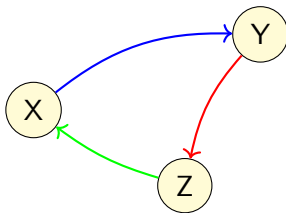
Original Polygon



Triangulated

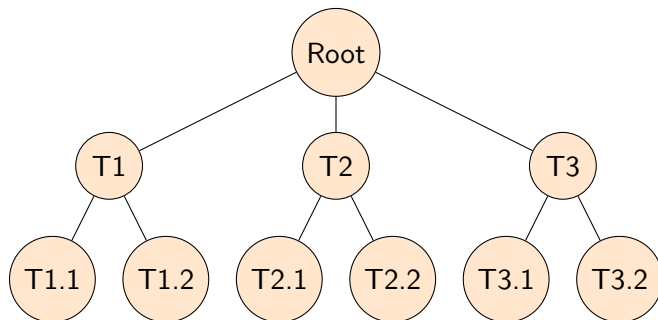


Non-straight edge example:



Algorithm Visualization

Tree structure of triangulations:



Graph with curved edges:

